Ohm's Law Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links voltage (V), current (I), and resistance (R). Cover the unknown to read the formula: $V = I \times R$, $I = V \div R$, and $R = V \div I$.

Round 1 — Retrieve It

1. What three quantities does the Ohm's law triangle connect?
2. Write the formula for voltage.
3. Write the formula for current.
4. Which variable do you cover to find resistance, and what formula results?

Round 2 — Sketch It

Draw the Ohm's law triangle. Put voltage on top; put current and resistance on the bottom. Circle the two quantities you would multiply to find voltage.

Ohm's Law — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A current of 2 A flows through a 6 Ω resistor. What is the voltage across it?
2. A 12 V battery is connected across a 4 Ω resistor. What is the current?
3. A voltage of 24 V drives a current of 3 A through a component. What is its resistance?

Round 4 — Reason from Evidence

A circuit keeps a steady voltage of 12 V. First a 3 Ω resistor is used, then it is replaced with a 6 Ω resistor. Use the triangle to find the current in each case, then say what happens to the current as resistance increases and how you know.

Ohm's Law — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the Ohm's law triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect Ohm's law to a simple circuit with a battery and a bulb. If the battery voltage stays fixed, explain why a bulb with higher resistance draws a lower current, using $I = V \div R$.

E

Extend

Create your own problem that asks for resistance. Give the numbers and show the work using $R = V \div I$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Ohm's Law

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Voltage, current, and resistance.
2. $V = I \times R$.
3. $I = V \div R$.
4. Cover R; $R = V \div I$.

Sketch It — look for

Voltage on top, with current and resistance on the bottom. Voltage is found by multiplying current × resistance (the two bottom quantities).

Apply It

1. $V = I \times R = 2 \times 6 =$ **12 V**.
2. $I = V \div R = 12 \div 4 =$ **3 A**.
3. $R = V \div I = 24 \div 3 =$ **8 Ω**.

Reason from Evidence — look for

3 Ω case: $I = 12 \div 3 = 4$ A. 6 Ω case: $I = 12 \div 6 = 2$ A. As resistance increases at constant voltage, the current decreases — an inverse relationship (doubling the resistance halves the current).

ACE — Extend look-for

Many answers possible. Check that the problem asks for resistance and the work correctly uses $R = V \div I$.

> **Suggested use.** Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.